

National Aeronautics
and Space Administration



Biomedical Foundation Models

Machine-learning and generative
models for mouse spaceflight RNA-seq

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Biological & Physical
Sciences



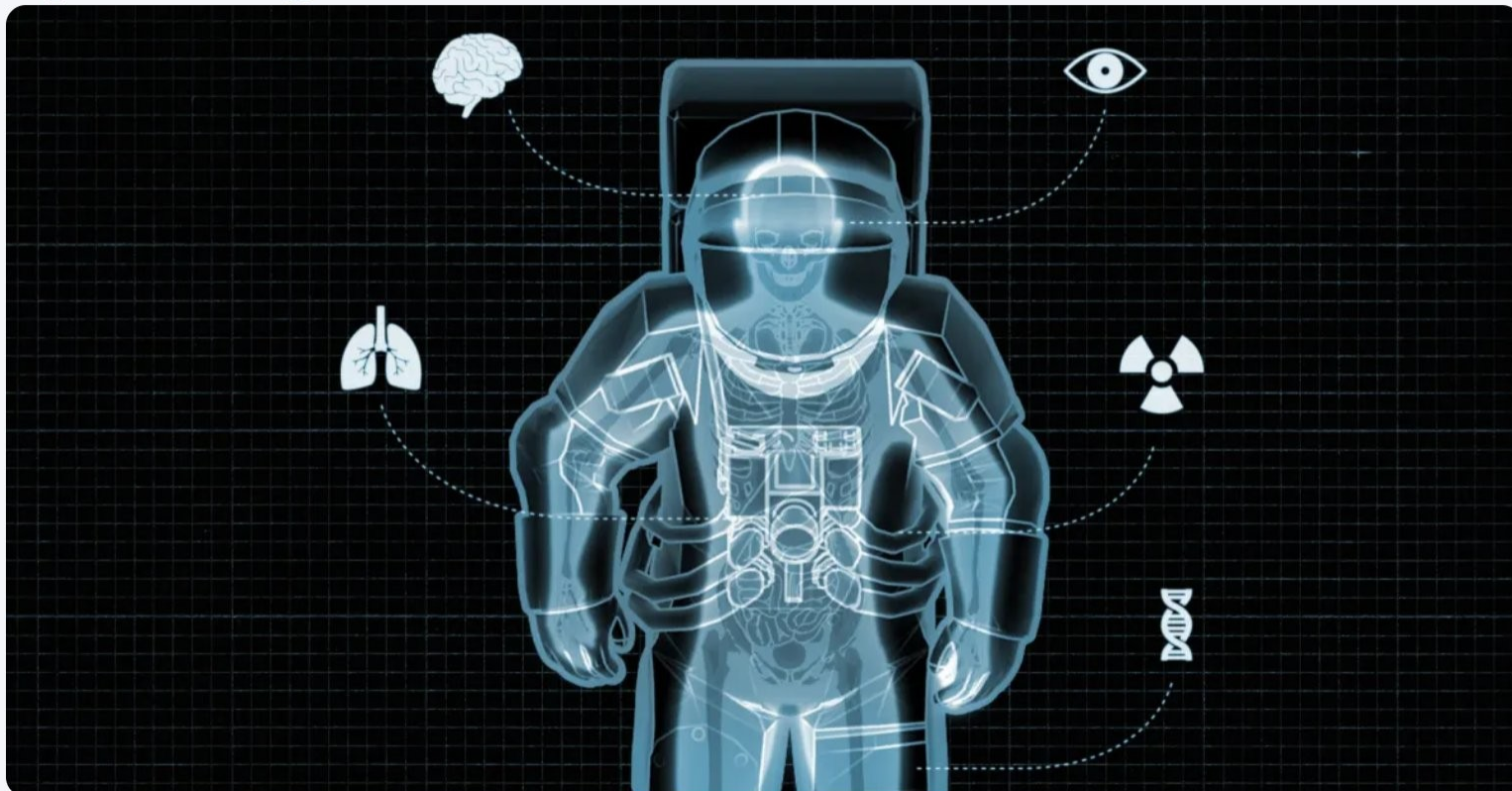
PROJECT OBJECTIVE

Learn how spaceflight changes living systems — and model the biology it produces

Using mouse bulk RNA-seq from NASA missions, we compare spaceflight (FLT) and ground-control (GC) mice across many tissues.

GOAL 1 · UNDERSTAND

Which genes, pathways, and systems does spaceflight alter?

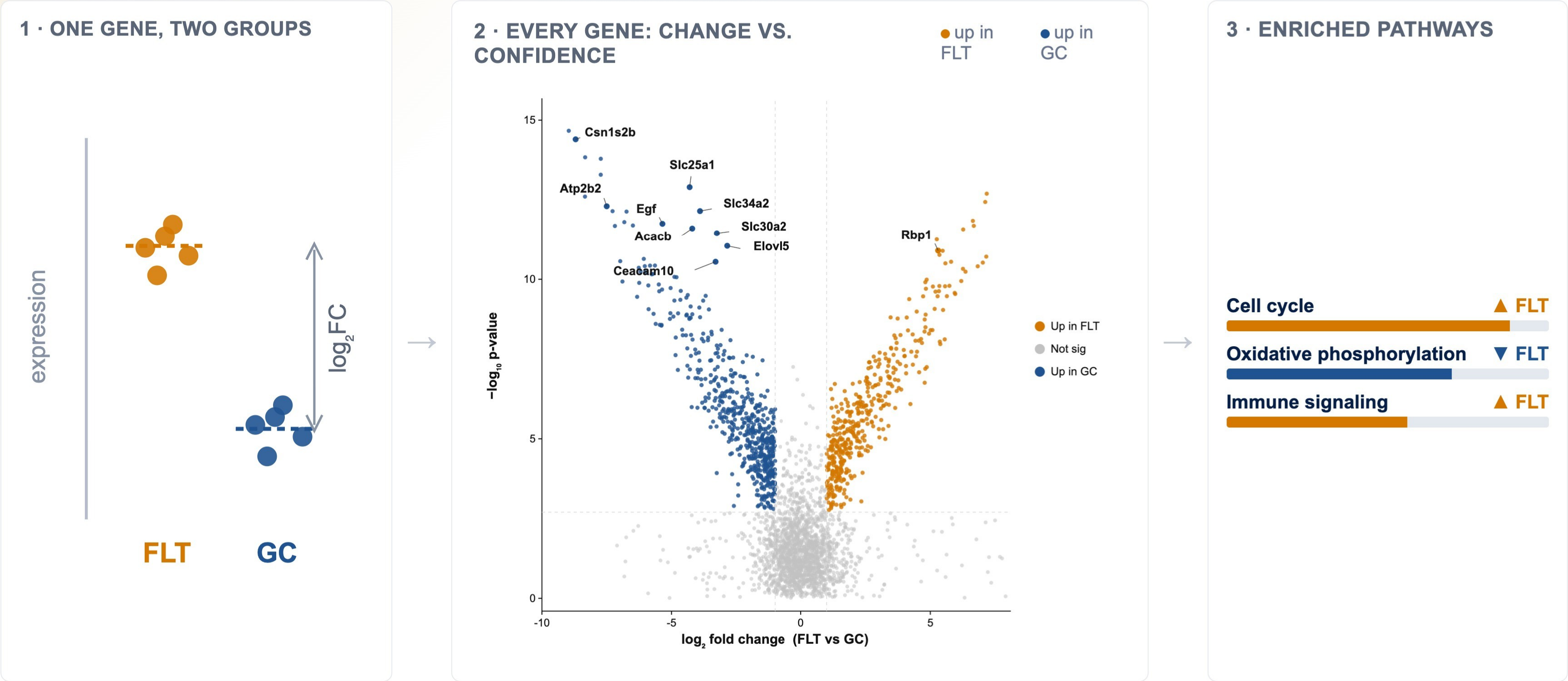
**GOAL 2 · MODEL & GENERATE**

Learn meaningful representations — and generate realistic profiles



THE BASELINE

Conventional differential expression, within each study



COMPLEMENTARY METHODS · CLASSICAL ML

Is there a repeatable flight signal? Yes.

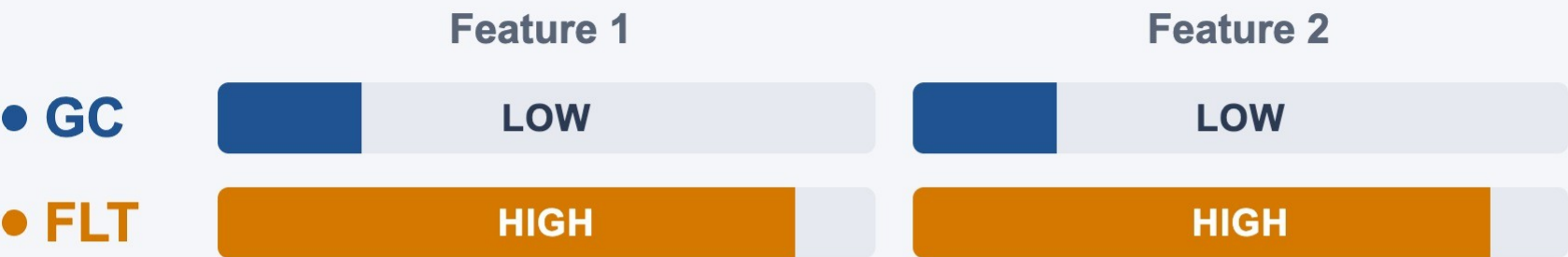
A classifier separates FLT from GC — a direct test of whether expression carries a learnable spaceflight signal.

HOW IT SEPARATES FLT FROM GC

● FLT ● GC



WHAT SEPARATES THEM

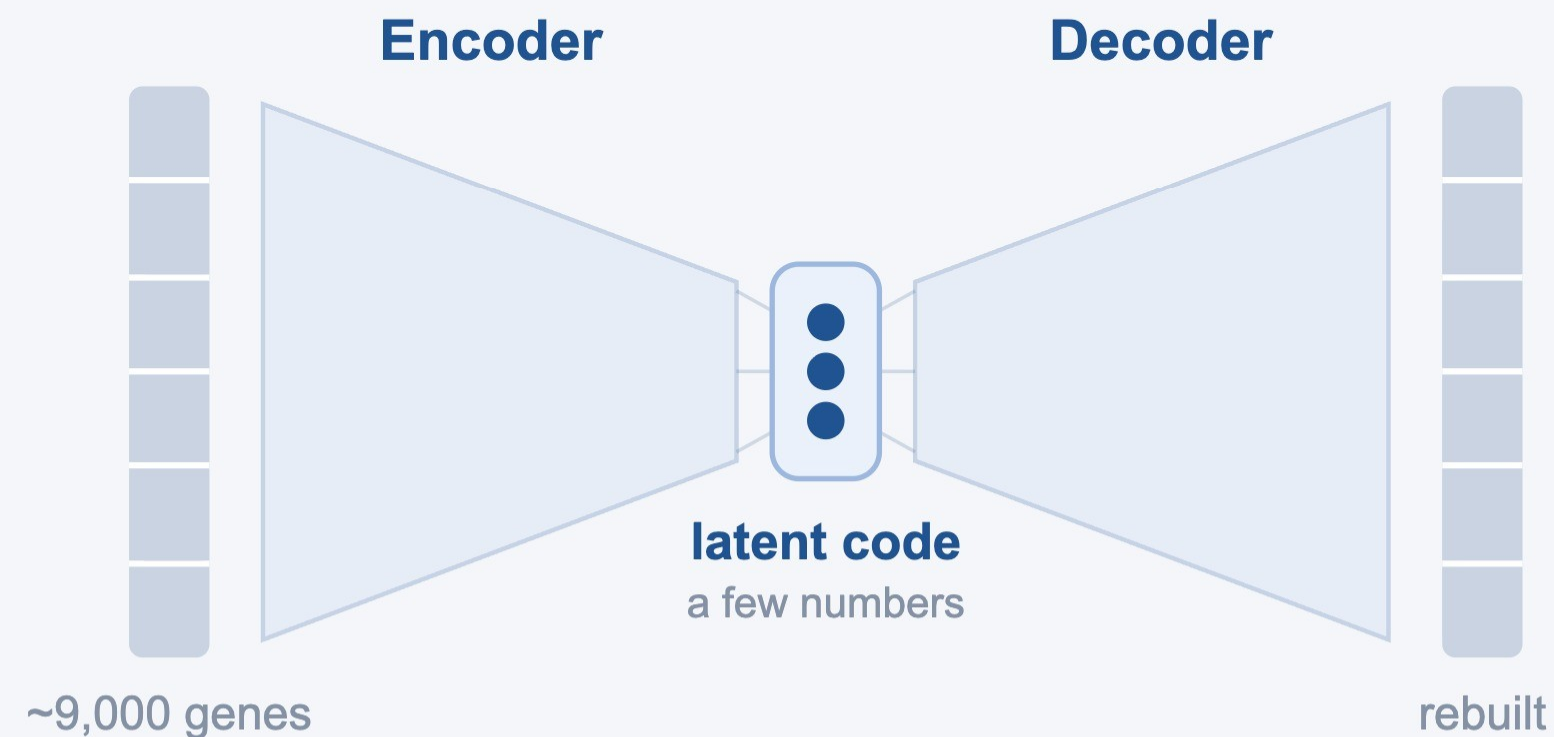


COMPLEMENTARY METHODS · NEURAL NETWORKS

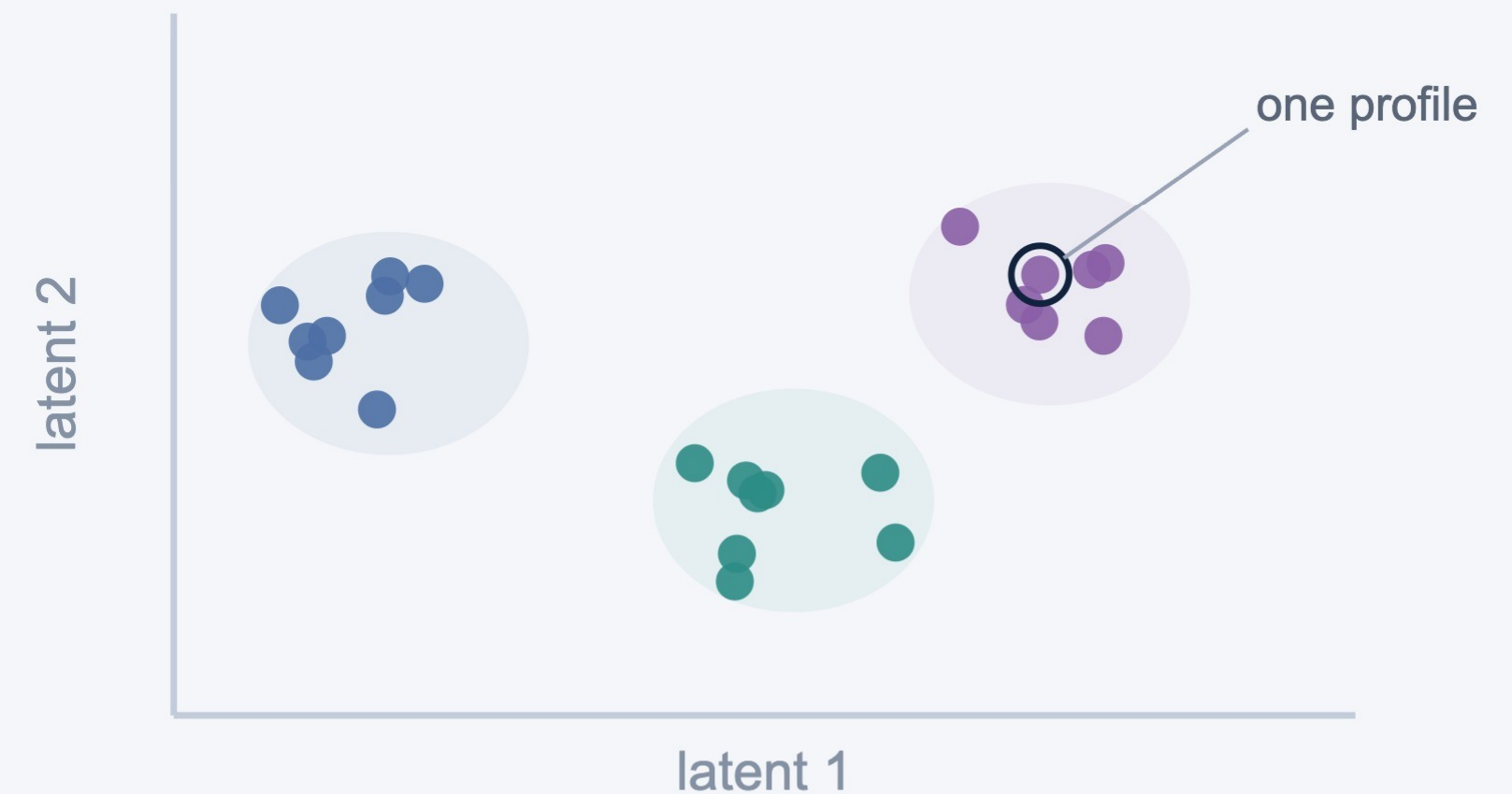
Can a model find hidden gene programs?

Autoencoder — squeeze, then rebuild

Compress every profile through a bottleneck, then reconstruct it.

**Latent space — a map of profiles**

Those few numbers place each profile as a point on a map.

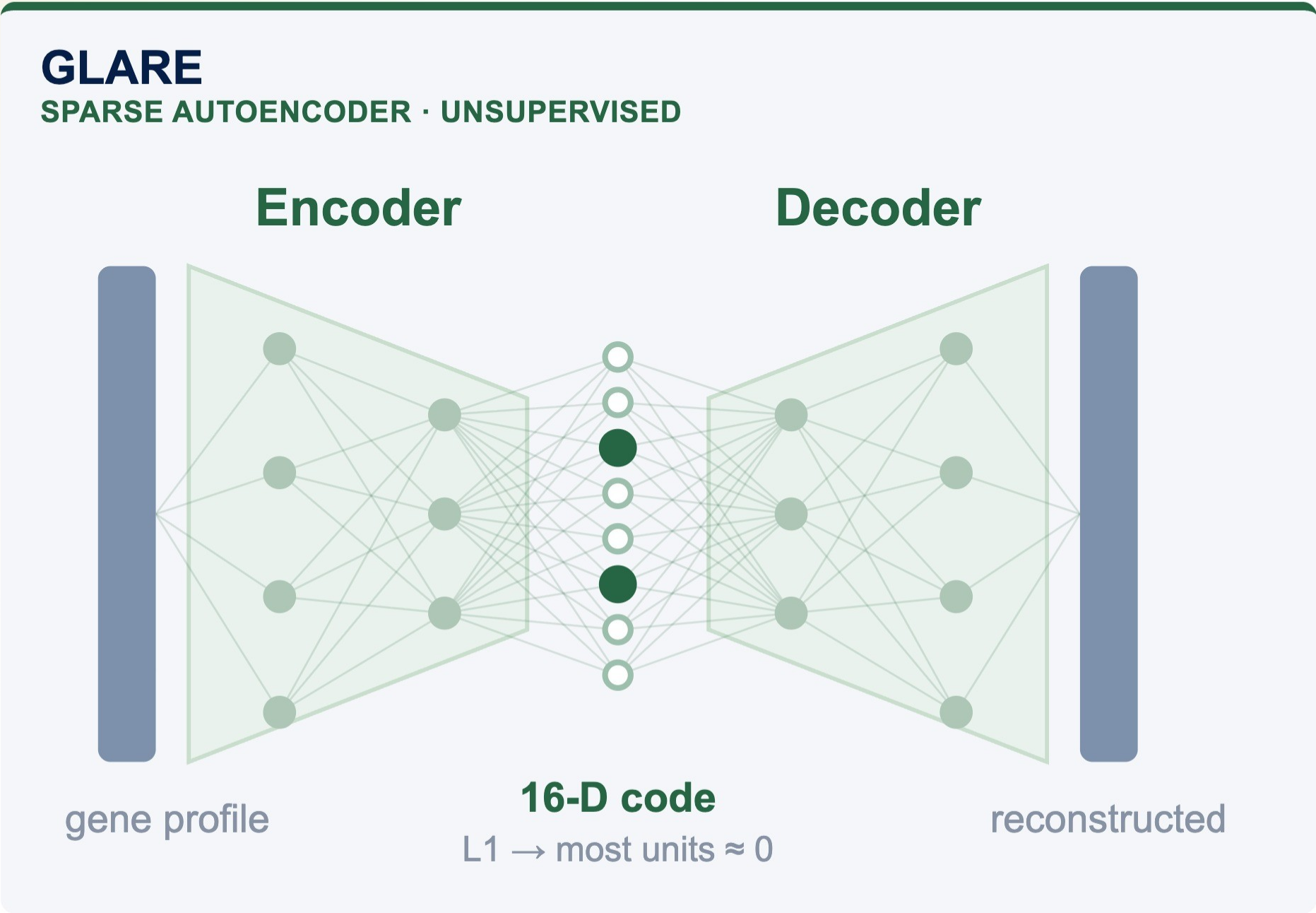


Latent space — a compressed, mathematical representation of complex data. It maps high-dimensional information — the ~9,000 genes in an expression profile, like the millions of pixels in an image — into a lower-dimensional space where similar profiles cluster close together.

NEURAL NETWORKS · OUR FIRST MODEL

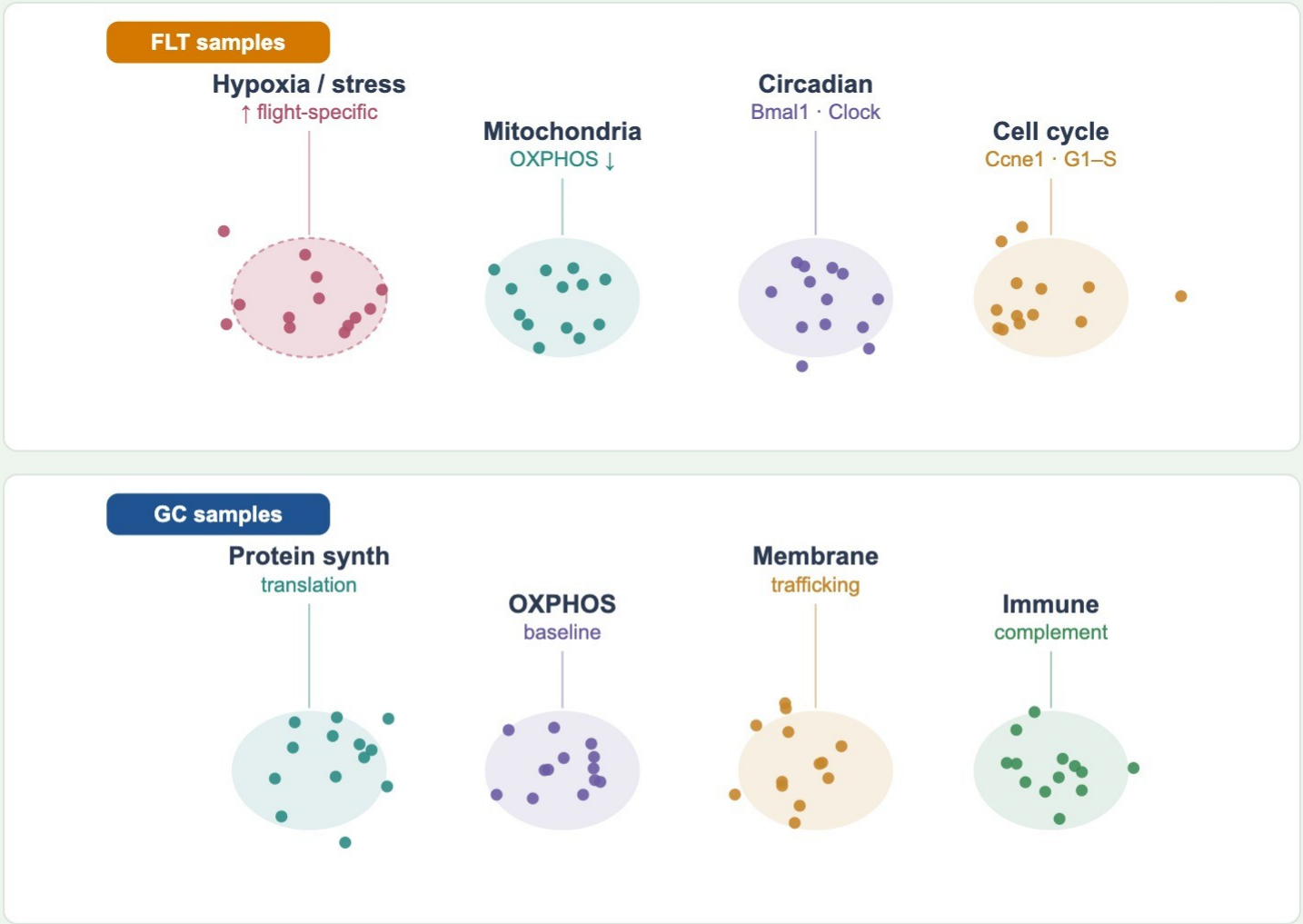
GLARE: an unsupervised search for gene programs

GLARE is a sparse autoencoder from the microgravity literature — it learns a latent code for each profile and groups genes into co-expression modules, without ever seeing the flight-versus-ground label.



THE GOAL

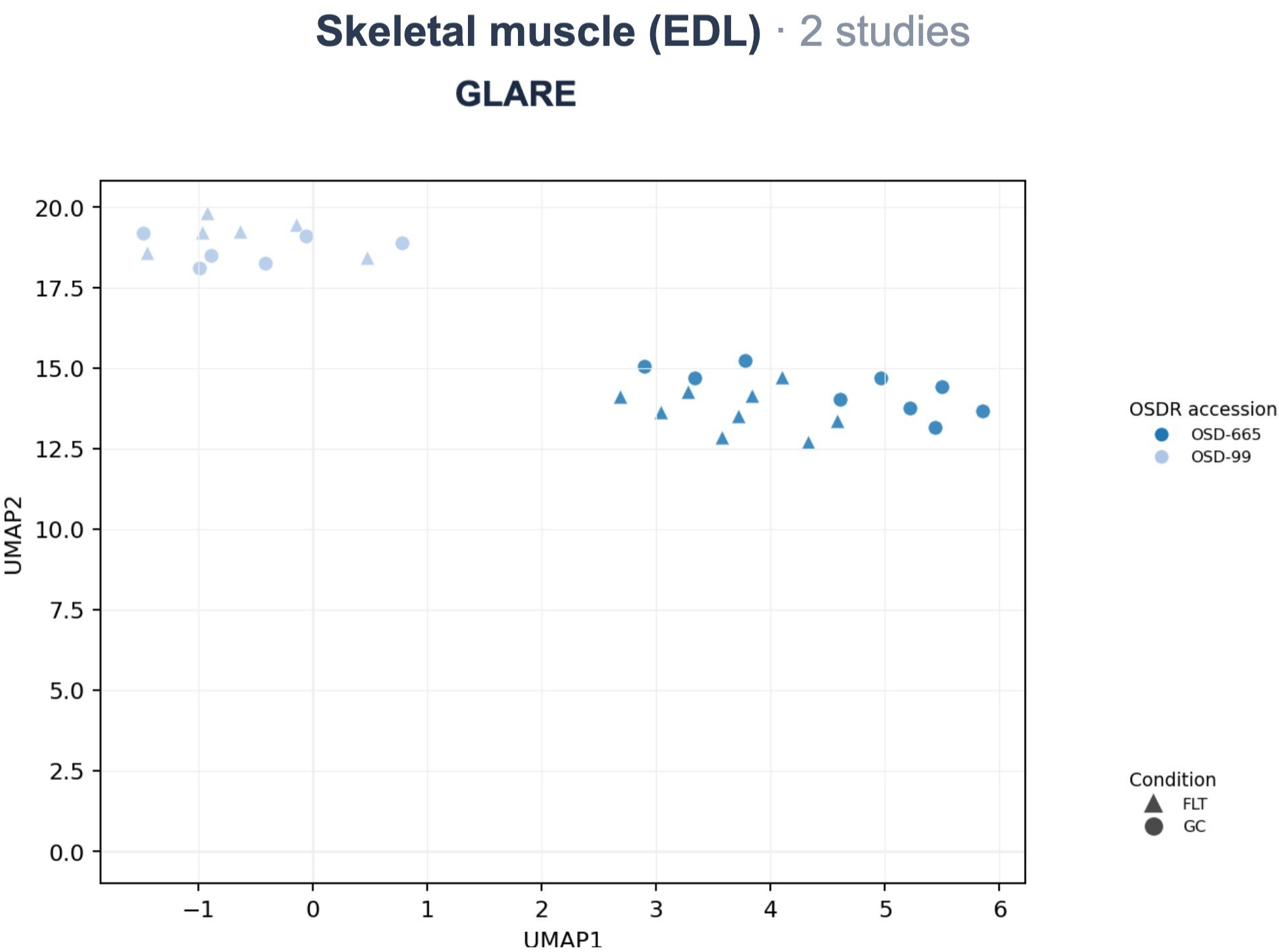
Encode each gene to a 16-D code, then cluster the genes into co-expression modules — separately for flight and ground.



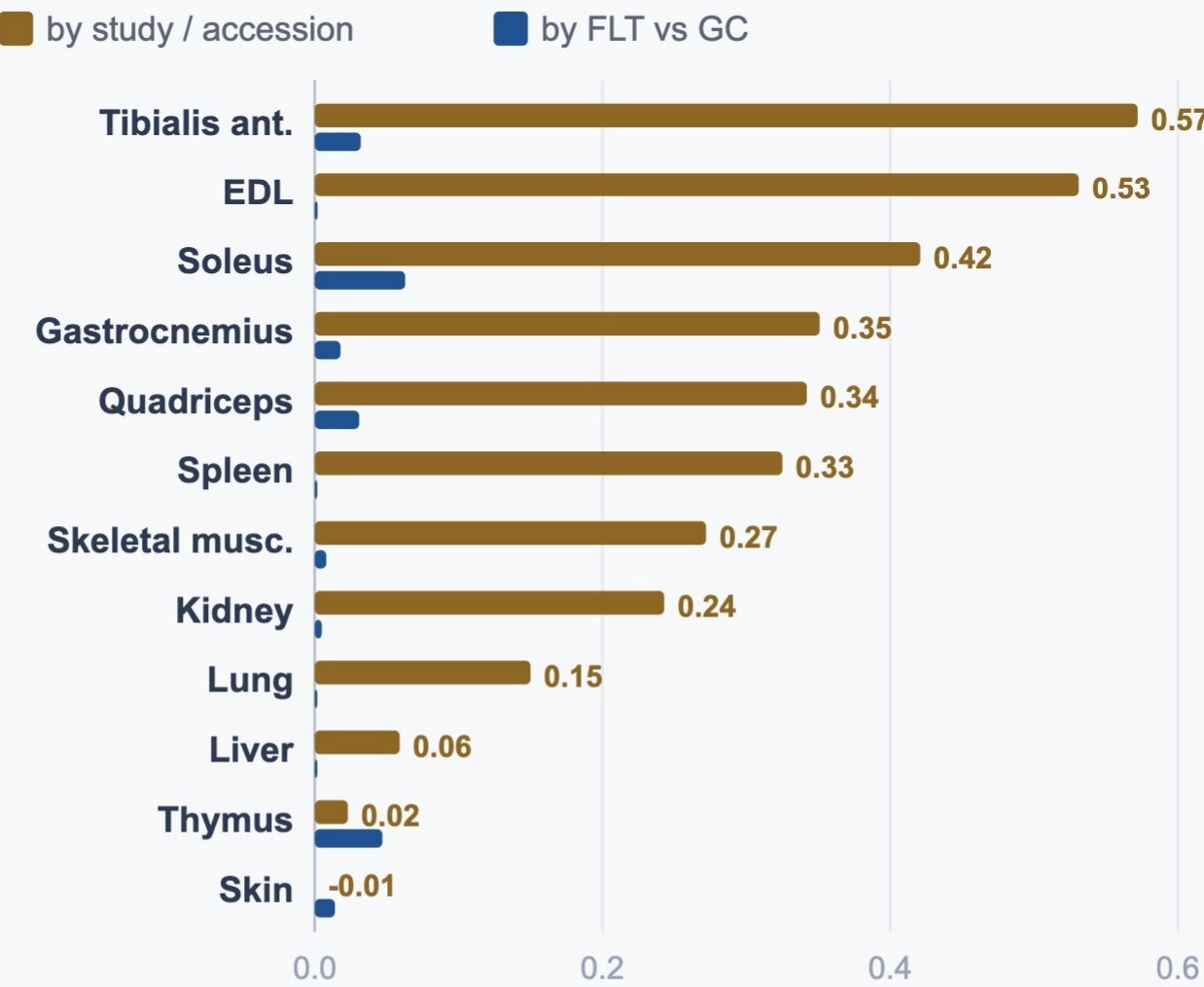
THE COMPLICATION

The biggest signal is the study — not the flight or the biology

Observed expression = **Spaceflight biology** + **Tissue biology** + **Animal variation** + **Study / protocol**



Silhouette per tissue · GLARE



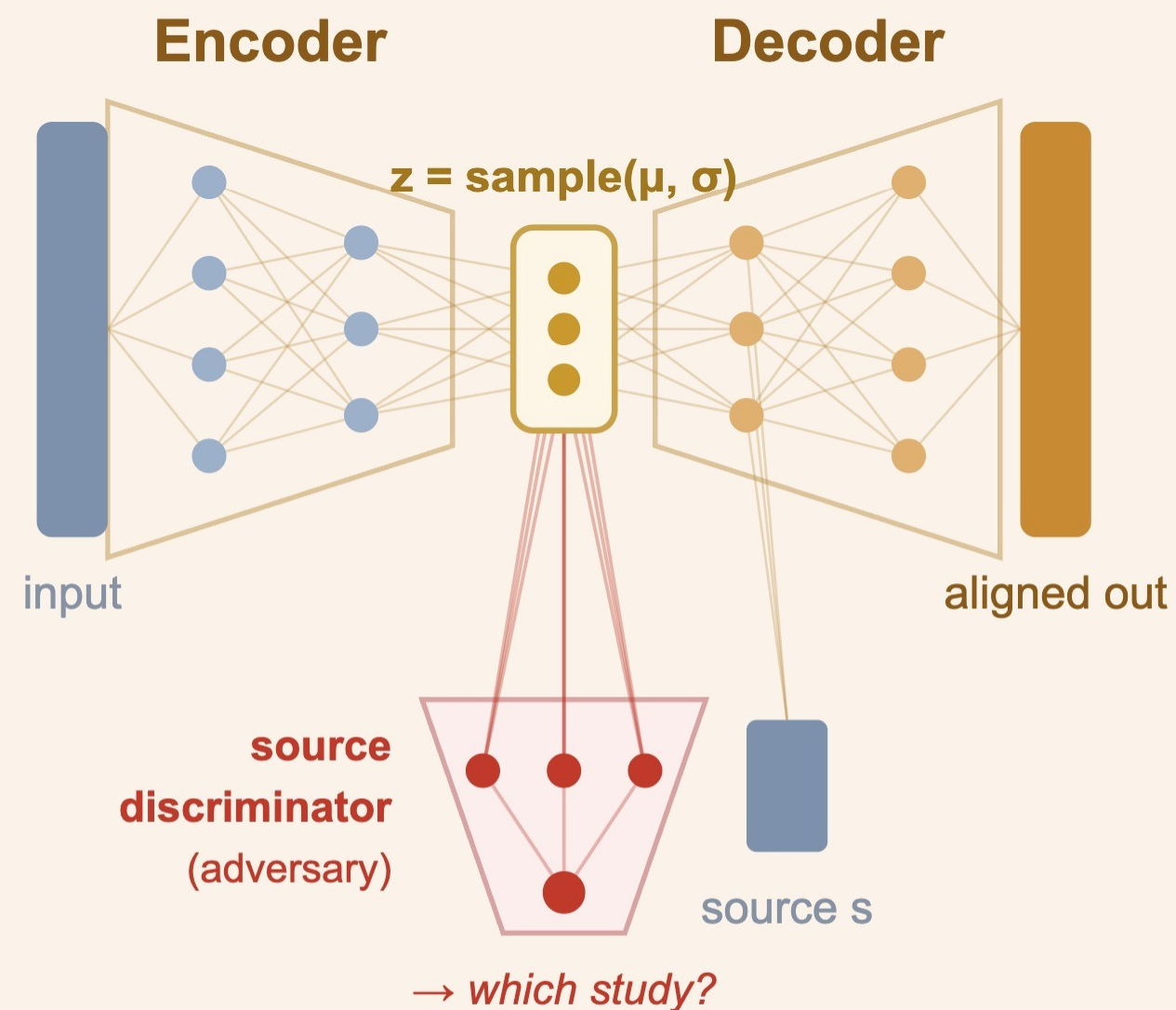
THE MAIN CHALLENGE

Can we just subtract the study effect?

MOBER (Multi-Origin Batch Effect Remover) is a deep domain-adaptation model that pulls studies into one distribution while trying to preserve biology.

How MOBER works

VAE encoder-decoder + source-discriminator adversary



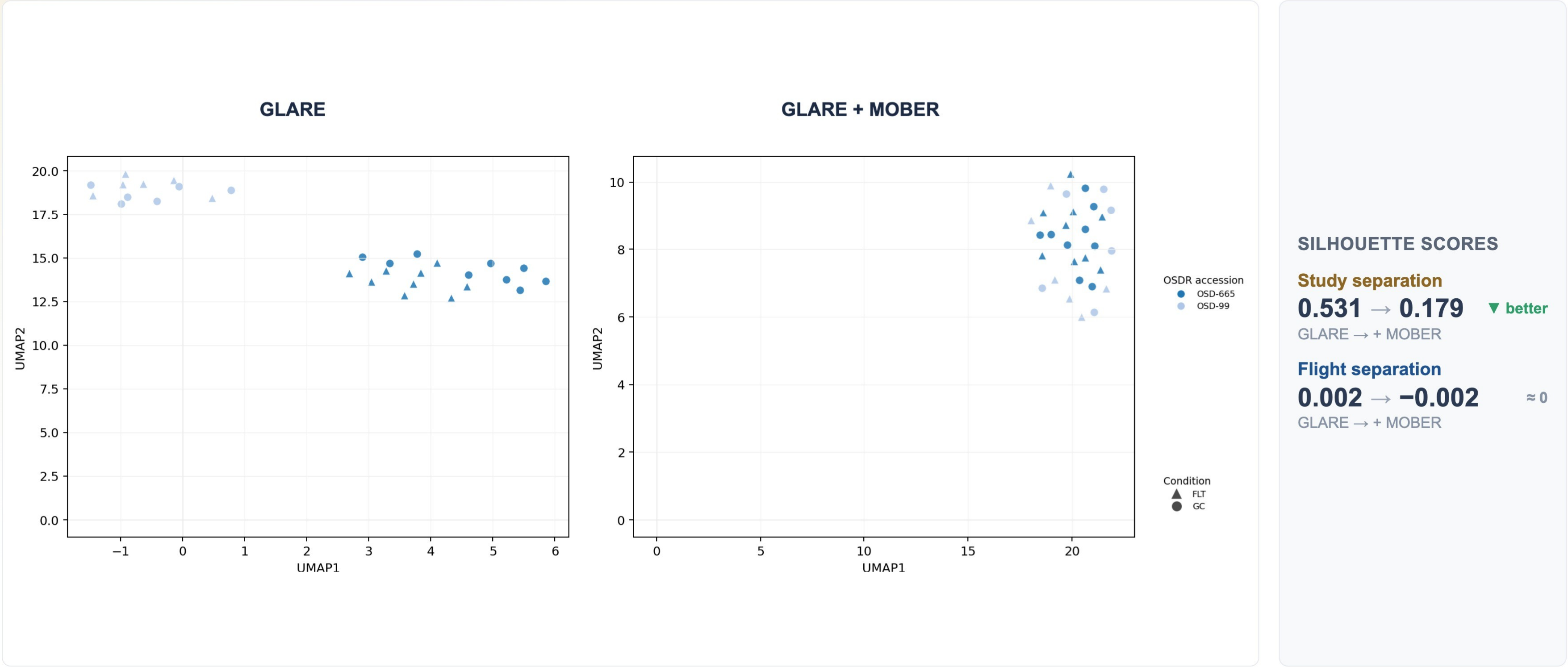
What it does · align studies

pull separate studies into one shared distribution

THE MAIN CHALLENGE · EVIDENCE

In skeletal muscle, MOBER helps — yet flight still doesn't emerge

Skeletal muscle (EDL): here MOBER **does** pull the two studies together — but FLT vs GC still doesn't separate, and the correction is only partial.



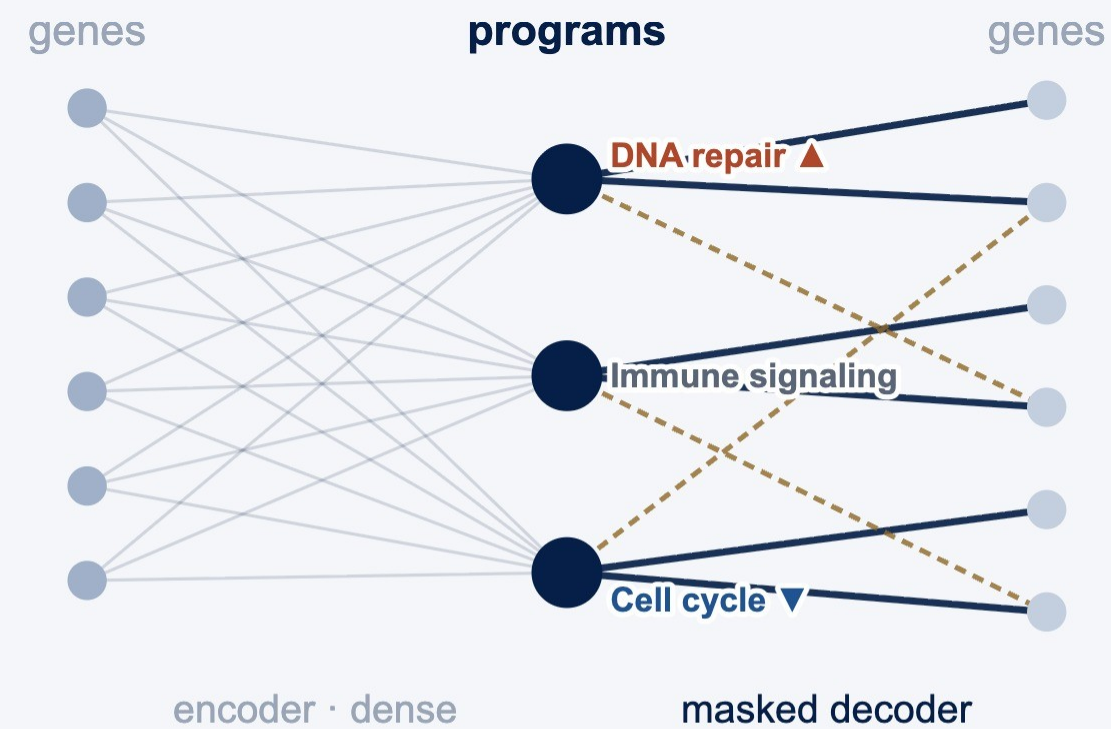
THE PIVOT · METHOD SPOTLIGHT

Don't learn the biology — wire it in

GLARE and MOBER tried to **learn** structure from the data. **expiMap** flips it — known Reactome pathways are **built into** the decoder, so each latent axis **is** biology by construction.

expiMap

STANDARD ENCODER · PATHWAY-MASKED DECODER



OUTPUT · PROGRAMS × SAMPLES



thymus · red ↑ / blue ↓ in FLT

WHY IT'S INTERPRETABLE

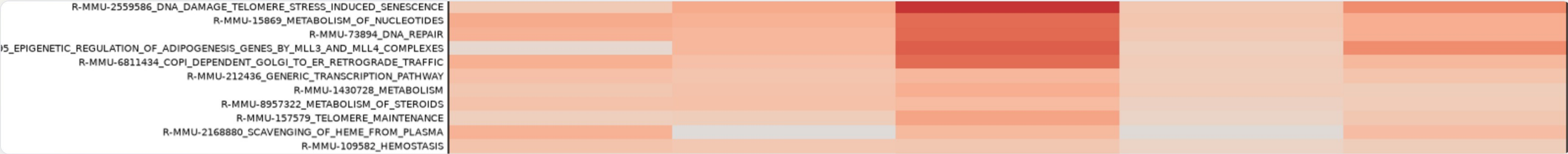
- Each latent node **is** one Reactome program (~1,140)
- A **soft mask** (dashed) lets a stray gene help only when the data demand it
- Compare **FLT** vs **GC** one program at a time

RESULTS · EXPIMAP · IS IT REAL?

① Pathway scores

Which pathway scores are real — and which are noise?

expiMap gives every program an activity score. Zooming into the top thymus programs — but a high score alone can't tell real biology from an artifact.



Thymus — 387 significant programs, top ones shown, ranked FLT → GC (red ↑ / blue ↓ in flight).

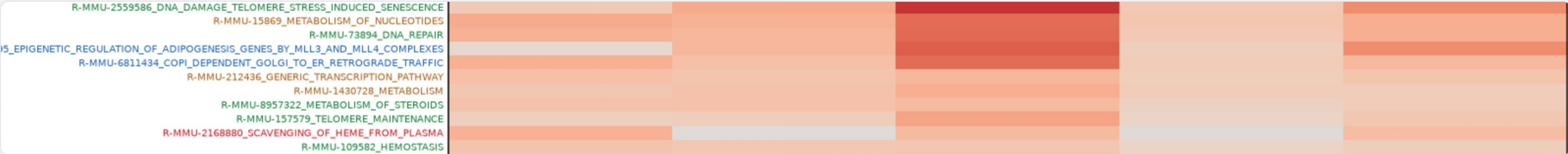
Modeled on the top-2,000 HVG to cut noise. Which are meaningful? ► review them

RESULTS · EXPIMAP · IS IT REAL?

② Literature-reviewed

Each program, reviewed against prior literature

We color-tag every program by how it compares to known spaceflight biology — then keep the meaningful ones and focus on the **complementary (blue)** programs.



COLORS ■ aligned ■ complementary ■ uncertain ■ conflict ■ low-effect

Thymus 387 → drop 217 gray · 24 red → 146 → 86 blue
· top-2,000 HVG

RESULTS · EXPIMAP · THYMUS

Thymus — 387 significant programs, reviewed

① All reviewed programs

RAW PATHWAY SCORES



An LLM reviews every significant program against prior spaceflight literature.



LABEL COLOR = LITERATURE FIT

- **aligned** — matches known biology
- **complementary** — new angle
- **uncertain**
- **conflict**
- **low-effect**

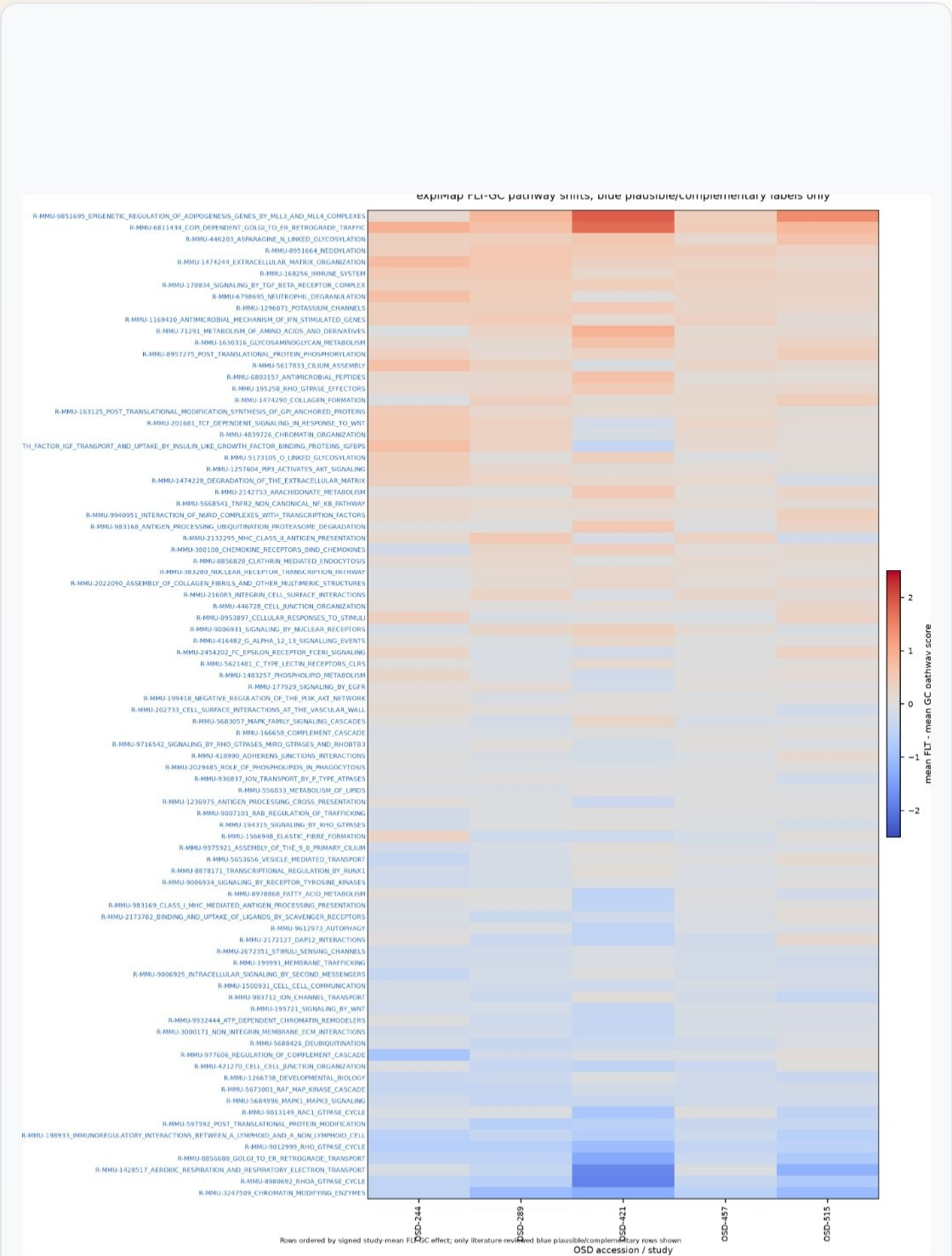
LITERATURE-REVIEWED



RESULTS · EXPIMAP · THYMUS

② Complementary signal · 86 programs

Thymus – involution & stromal-niche remodeling



① KNOWN PHENOTYPE · PRIOR LITERATURE

Spaceflight causes thymic involution, reduced thymopoiesis, impaired T-cell activation, apoptosis and glucocorticoid stress, and innate-immune remodeling.

② ALIGNED PROGRAMS · RECOVER KNOWN BIOLOGY

Cell cycle, CD28 / T-cell activation and interleukin signaling down; DNA repair, telomere stress, cell death, steroid metabolism and innate / TLR / hemostasis up.

Examples: Cell Cycle ↓, T-cell Activation (CD28) ↓, DNA Repair ↑, Programmed Cell Death ↑, Steroid Metabolism ↑, Hemostasis ↑

③ COMPLEMENTARY PROGRAMS · NEW HYPOTHESES

Thymic-niche communication falls ↓ lower in flight

Rho / RAC cytoskeletal, junction and lymphoid–stromal interaction scores are lower — impaired thymocyte migration and stromal crosstalk.

Example: Rho/RAC GTPase cycle ↓, cell–cell junctions ↓, lymphoid–non-lymphoid interactions ↓

Stromal matrix remodeling rises ↑ higher in flight

ECM, collagen, TGF-β and WNT/TCF pathways are higher — the thymic scaffold actively remodels during involution.

Example: ECM organization ↑, collagen formation ↑, TGF-β ↑, TCF/WNT ↑

Innate immune stress switches on ↑ higher in flight

Innate / antiviral and inflammatory pathways rise alongside the adaptive-immune suppression.

Example: IFN antimicrobial ↑, neutrophil degranulation ↑, TNFR2/NF-κB ↑

Respiratory / activation metabolism falls ↓ lower in flight

Respiratory electron transport and MAPK / RAF signaling are lower — reduced mitochondrial and growth-state signaling.

Example: aerobic respiration ↓, MAPK1/3 ↓, RAF-MAPK ↓

Regulatory survival programs rise ↑ higher in flight

Chromatin organization and PIP3–AKT are higher — compensatory regulatory and survival / stress-response programming.

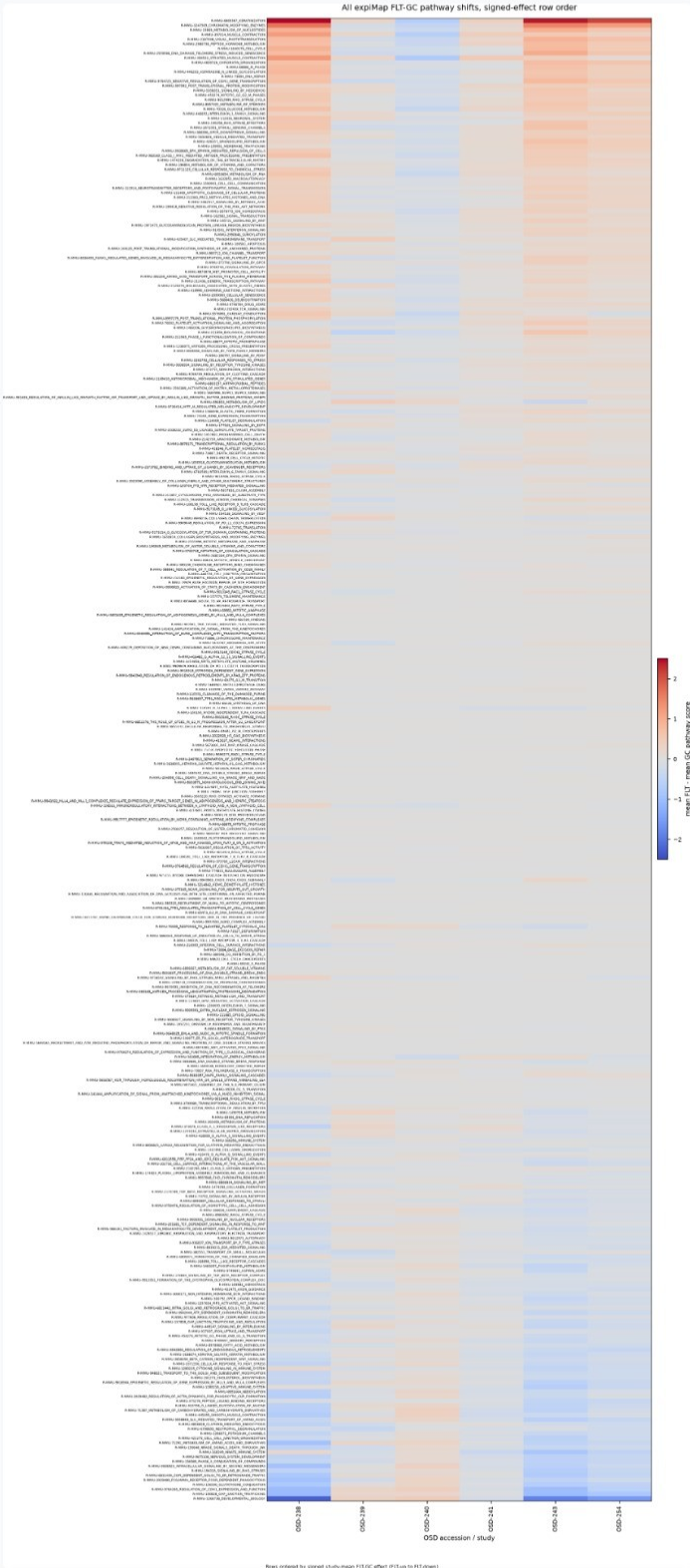
Example: chromatin organization ↑, PIP3–AKT ↑

RESULTS · EXPIMAP · SKIN

Skin — 319 significant programs, reviewed

① All reviewed programs

RAW PATHWAY SCORES



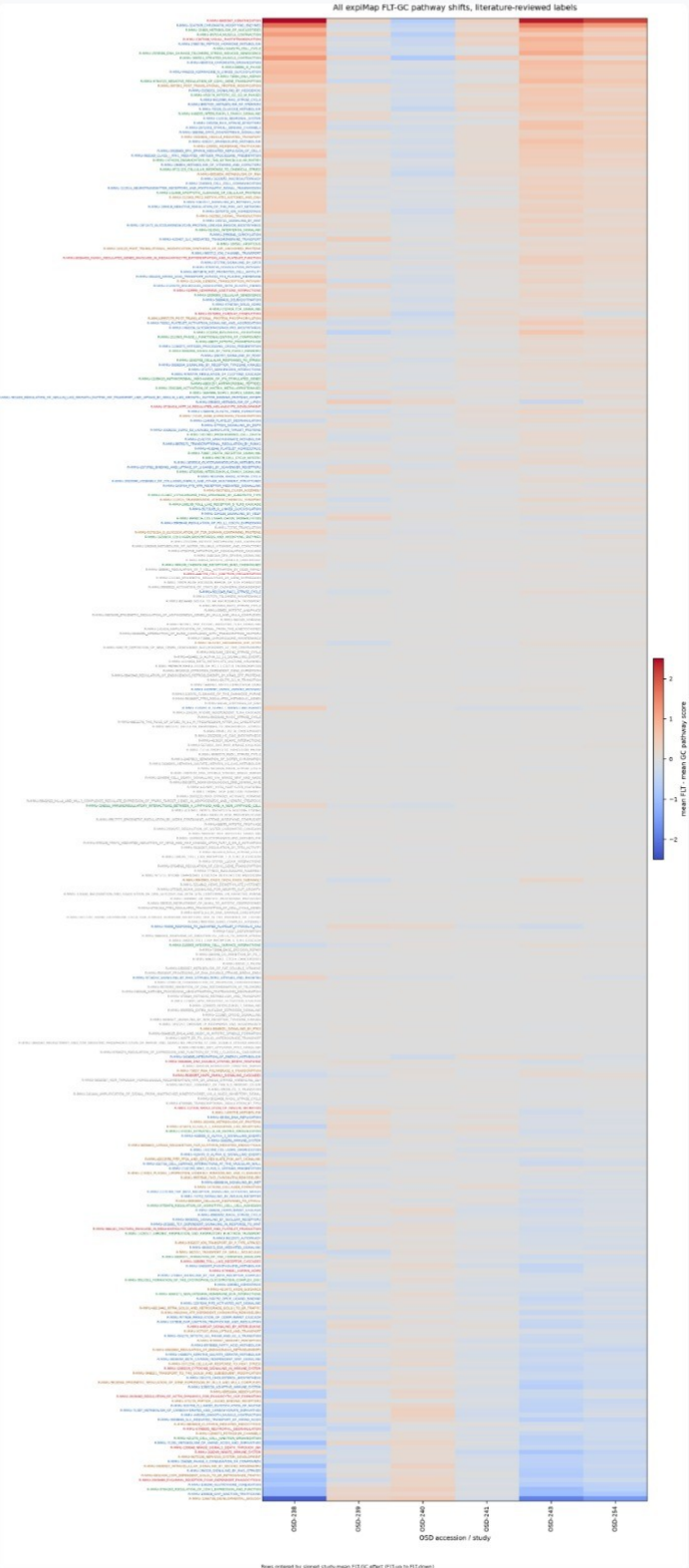
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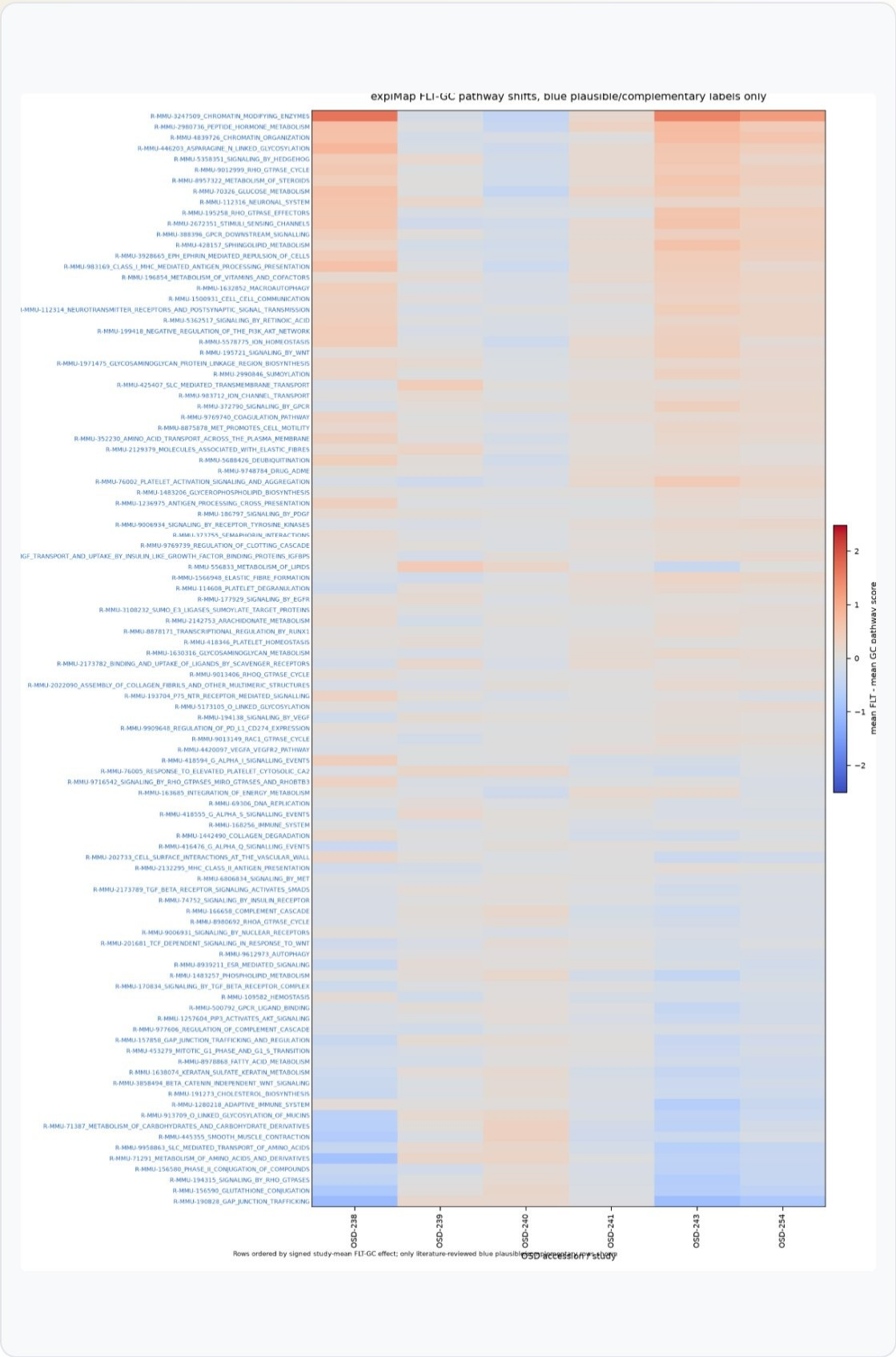
LITERATURE-REVIEWED



RESULTS · EXPIMAP · SKIN

② Complementary signal · 102 programs

Skin — repair signaling rises as barrier coordination weakens



● ① KNOWN PHENOTYPE · PRIOR LITERATURE

Spaceflight skin shows barrier disruption, keratin / cornification changes, inflammation, DNA damage, oxidative stress, dermal / ECM remodeling and wound-healing risk.

■ ② ALIGNED PROGRAMS · RECOVER KNOWN BIOLOGY

DNA repair, cell cycle, IL-1 / interferon / antimicrobial immunity and ECM degradation up; cell-junction, adhesion and cornified-envelope programs down.

Examples: DNA Repair ↑, Cell Cycle ↑, Interleukin-1 ↑, Interferon ↑, ECM Degradation ↑, Cornified Envelope ↓

■ ③ COMPLEMENTARY PROGRAMS · NEW HYPOTHESES

Repair signaling rises ↑ higher in flight

Growth-factor and migration pathways are higher — wound-repair and regenerative signaling beyond inflammation.

Example: WNT ↑, Hedgehog ↑, RTK ↑, GPCR ↑, Rho effectors ↑

Cell–cell coordination falls ↓ lower in flight

Gap-junction trafficking is lower — weaker intercellular coordination during barrier recovery.

Example: gap-junction trafficking ↓

Cell-autonomous stress control rises ↑ higher in flight

Chromatin remodeling, SUMOylation and autophagy are higher — stress control, damage response and transcriptional adaptation.

Example: chromatin-modifying enzymes ↑, SUMOylation ↑, macroautophagy ↑

Detox / antioxidant capacity falls ↓ lower in flight

Glutathione and phase-II conjugation are lower — reduced detoxification and antioxidant-conjugation capacity.

Example: glutathione conjugation ↓, phase-II conjugation ↓

Barrier-lipid remodeling rises ↑ higher in flight

Sphingolipid metabolism is higher — active remodeling of epidermal barrier-lipid programs.

Example: sphingolipid metabolism ↑

Barrier metabolic support falls ↓ lower in flight

Cholesterol biosynthesis and fatty-acid metabolism are lower — reduced lipid-metabolic support for barrier maintenance.

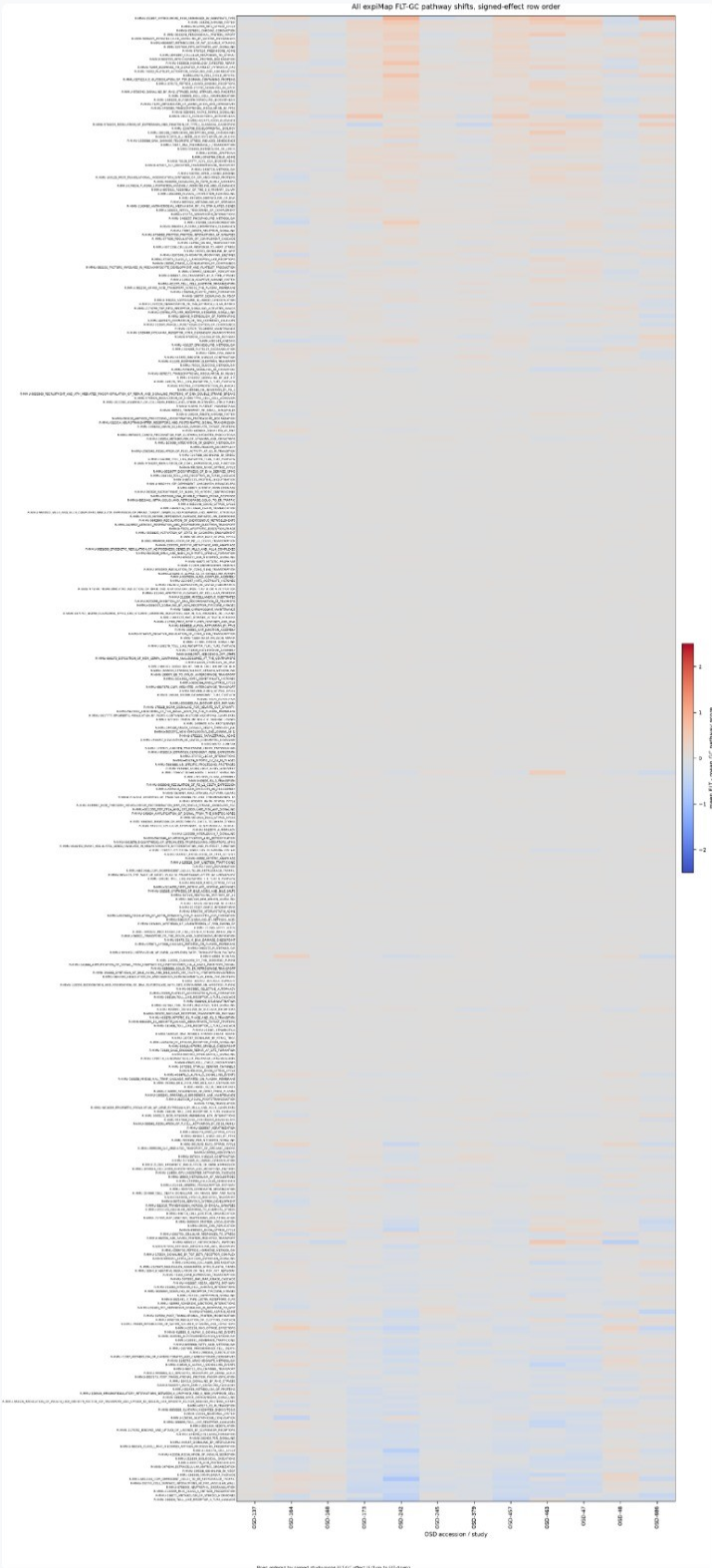
Example: cholesterol biosynthesis ↓, fatty-acid metabolism ↓

RESULTS · EXPIMAP · LIVER

Liver — 364 significant programs, reviewed

① All reviewed programs

RAW PATHWAY SCORES



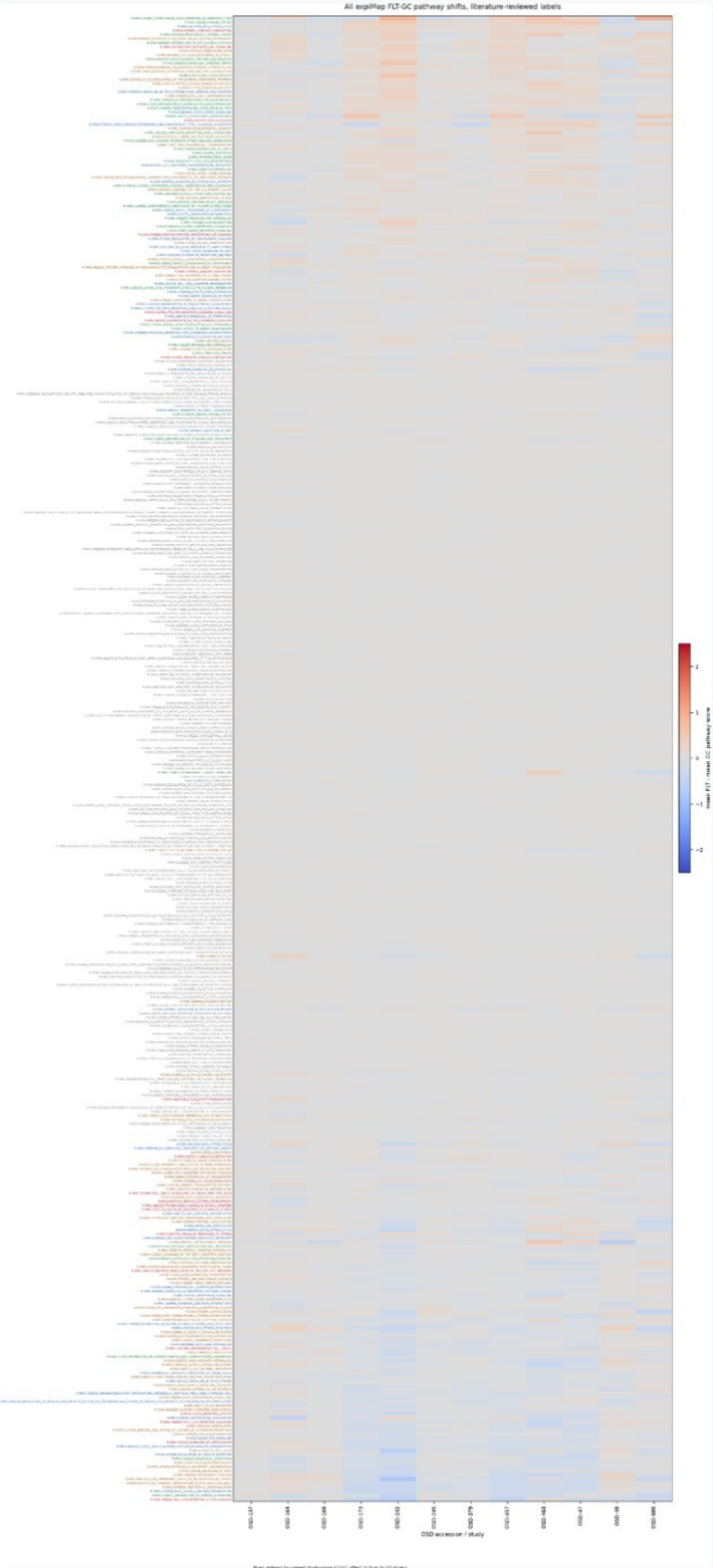
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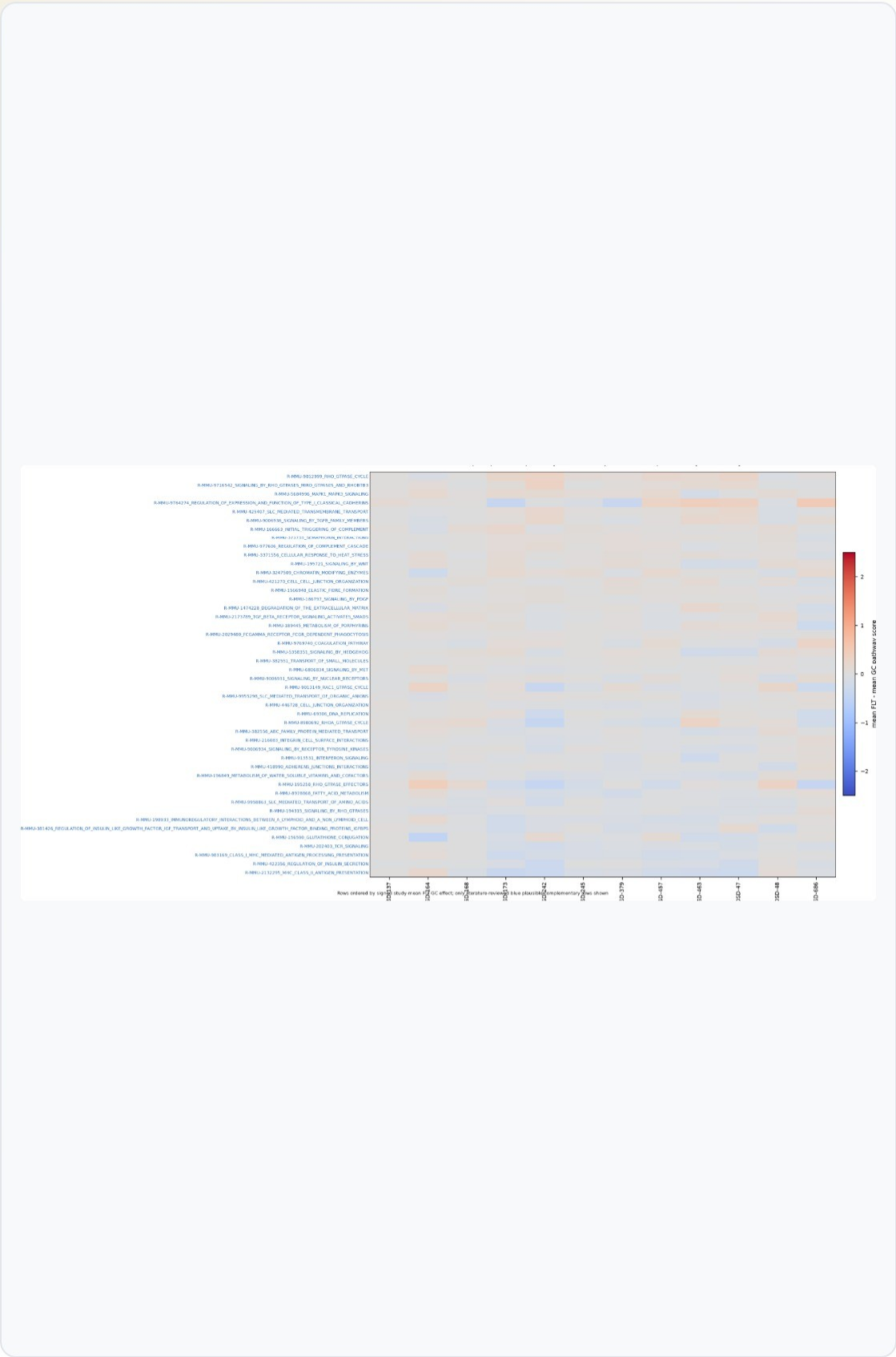
LITERATURE-REVIEWED



RESULTS · EXPIMAP · LIVER

② Complementary signal · 46 programs

Liver — metabolic injury & mechanotransduction remodeling



① KNOWN PHENOTYPE · PRIOR LITERATURE

Spaceflight disrupts hepatic lipid handling, CYP / drug metabolism, carbohydrate and endocrine signaling, oxidative stress and immune balance.

② ALIGNED PROGRAMS · RECOVER KNOWN BIOLOGY

Lipid & lipoprotein metabolism, CYP / drug ADME, immune signaling and DNA-repair / p53 up; carbohydrate metabolism and biological oxidations down.
Examples: Cytochrome P450 ↑, Drug ADME ↑, Plasma Lipoprotein Remodeling ↑, DNA Repair ↑, TP53 ↑, Apoptosis ↑

③ COMPLEMENTARY PROGRAMS · NEW HYPOTHESES

Liver senses mechanical load differently ↑ higher in flight
Rho / MIRO cytoskeletal programs are higher — altered mechanotransduction and mitochondrial–cytoskeletal stress.
Example: RHO GTPase cycle ↑, RHO/MIRO signaling ↑

Tissue-remodeling signals switch on ↑ higher in flight
Matrix turnover and growth-factor pathways rise — repair and remodeling downstream of hepatic stress.
Example: ECM degradation ↑, TGF-β ↑, PDGF ↑, MET ↑

Adaptive immune activity is turned down ↓ lower in flight
Antigen presentation and T-cell signaling drop — immune suppression or cell redistribution, not simple inflammation.
Example: MHC I/II ↓, TCR signaling ↓, interferon ↓

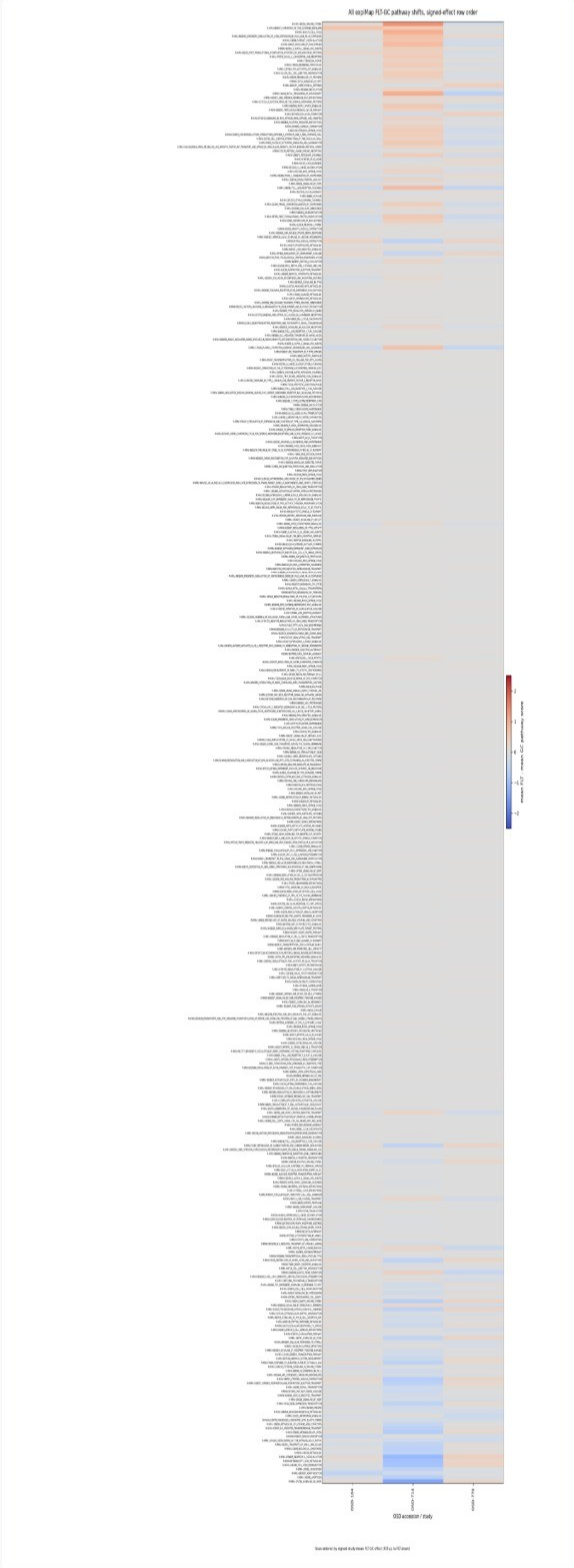
Metabolic support is depleted ↓ lower in flight
Fatty-acid, amino-acid transport, vitamin / cofactor and glutathione pathways drop — suppressed metabolic and antioxidant support.
Example: fatty-acid metabolism ↓, amino-acid transport ↓, vitamins/cofactors ↓, glutathione ↓

RESULTS · EXPIMAP · SOLEUS

Soleus — 357 significant programs, reviewed

① All reviewed programs

RAW PATHWAY SCORES



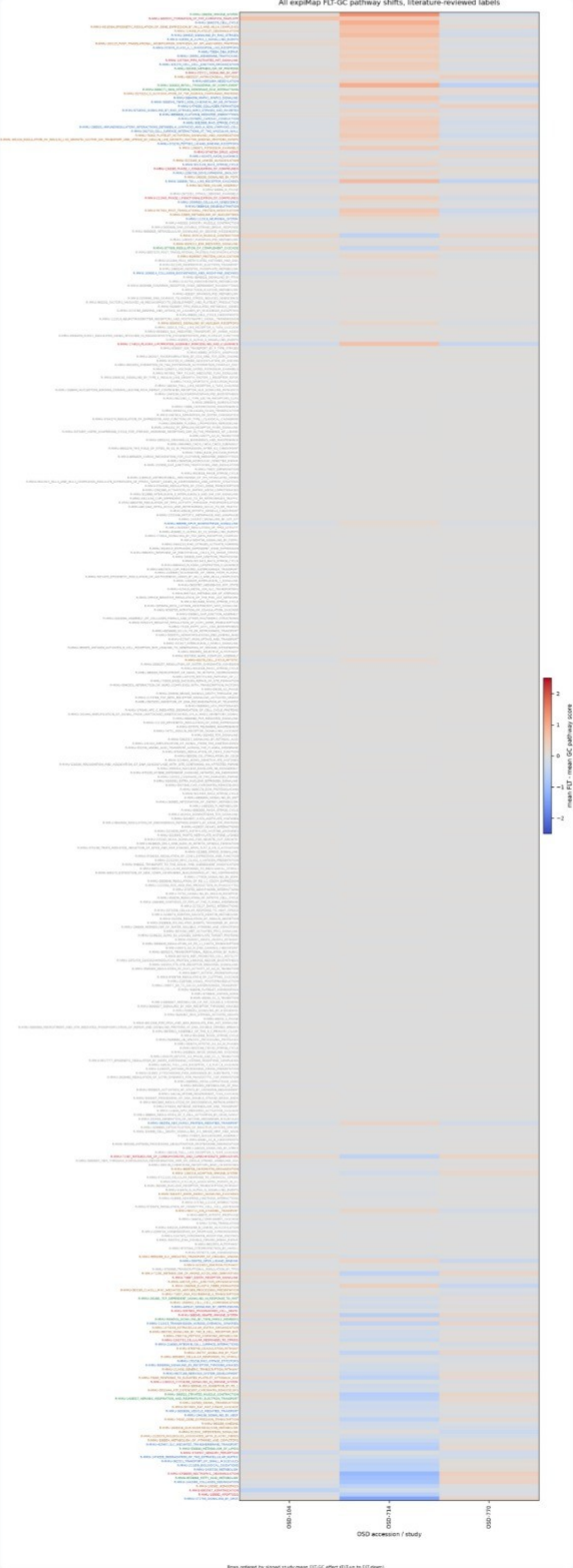
An LLM reviews every significant program against prior spaceflight literature.



LABEL COLOR = LITERATURE FIT

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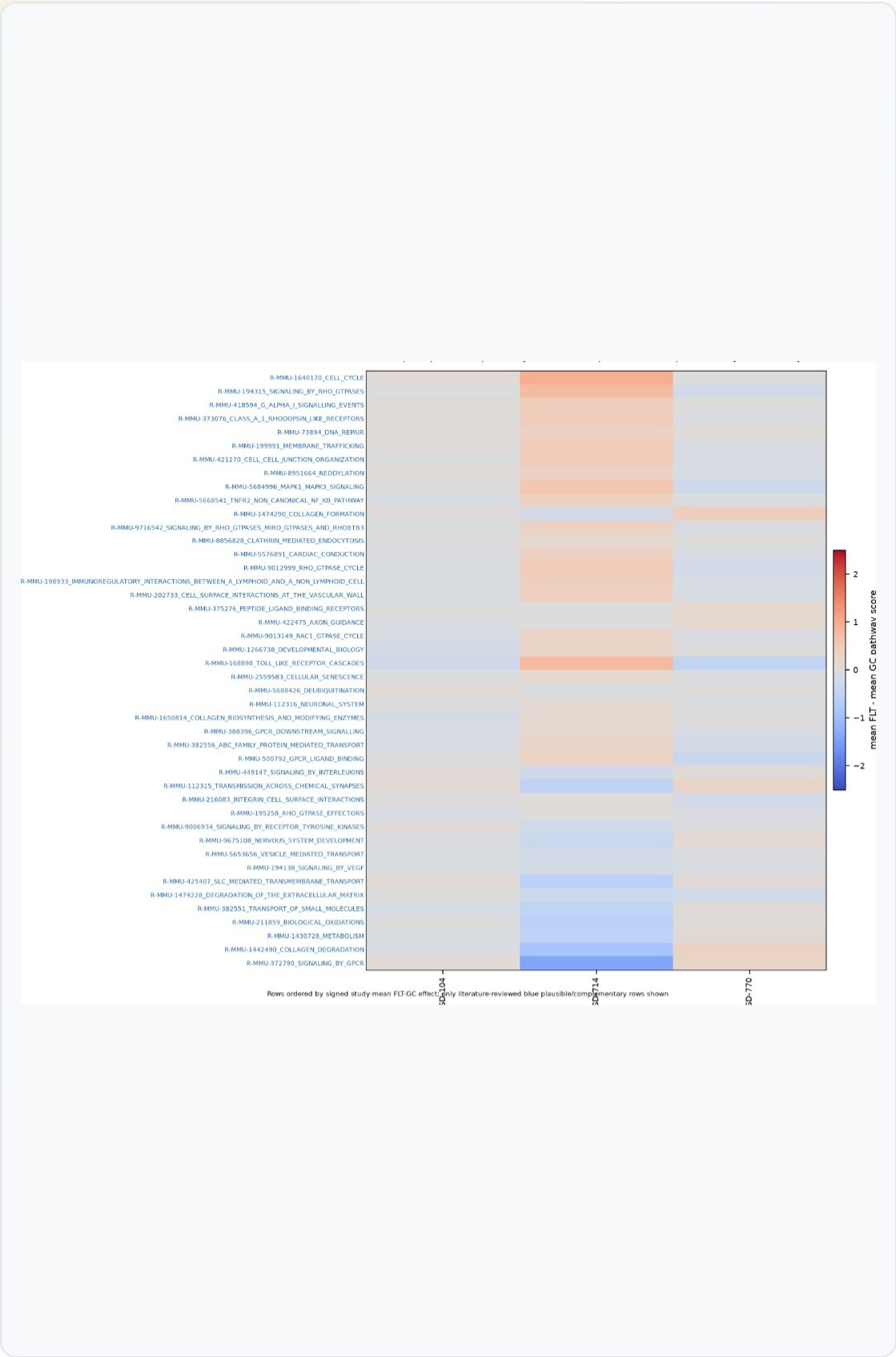
LITERATURE-REVIEWED



RESULTS · EXPIMAP · SOLEUS

② Complementary signal · 44 programs

Soleus — atrophy plus matrix & load-sensing remodeling



● ① KNOWN PHENOTYPE · PRIOR LITERATURE

Spaceflight causes soleus atrophy, reduced contractile / oxidative metabolism, lipid suppression, ECM / collagen remodeling, inflammation and complement activation.

■ ② ALIGNED PROGRAMS · RECOVER KNOWN BIOLOGY

Fatty-acid & lipid metabolism, aerobic respiration and muscle contraction down; immune, complement, protein metabolism and ECM interactions up.

Examples: Fatty Acid Metabolism ↓, Aerobic Respiration ↓, Striated Muscle Contraction ↓, Complement ↑, ECM Interactions ↑

■ ③ COMPLEMENTARY PROGRAMS · NEW HYPOTHESES

Metabolic & transport capacity falls ↓ lower in flight

Broad metabolism, biological oxidations and transporters are lower — atrophy extends into metabolite / ion-transport loss.

Example: metabolism ↓, biological oxidations ↓, small-molecule & SLC transport ↓

Matrix breakdown falls ↓ lower in flight

Collagen and ECM degradation pathways are lower — altered matrix turnover, possibly reduced breakdown during remodeling.

Example: collagen degradation ↓, ECM degradation ↓

Load-sensing repair signals rise ↑ higher in flight

Rho, RAC, MAPK, junction and NF-κB pathways are higher — mechanotransduction and inflammatory remodeling layered on atrophy.

Example: RHO ↑, RAC1 cycle ↑, cell–cell junctions ↑, MAPK1/3 ↑, TNFR2/NF-κB ↑

Myogenic / nonmyofiber activity appears ↑ higher in flight

Cell-cycle and developmental programs are higher — repair / reprogramming or nonmyofiber activity, not mature muscle growth.

Example: cell cycle ↑, developmental biology ↑, DNA repair ↑

NEXT PHASE · GENERATIVE MODELS**Learning to generate realistic RNA-seq**

Few samples limit deep models; a generator that learns the data distribution could augment, simulate, and generate hypotheses.

GeneJepa**Conditional WGAN-GP****Diffusion · DDPM/DDIM**

HOW THE GENERATORS WORK · 1 OF 3

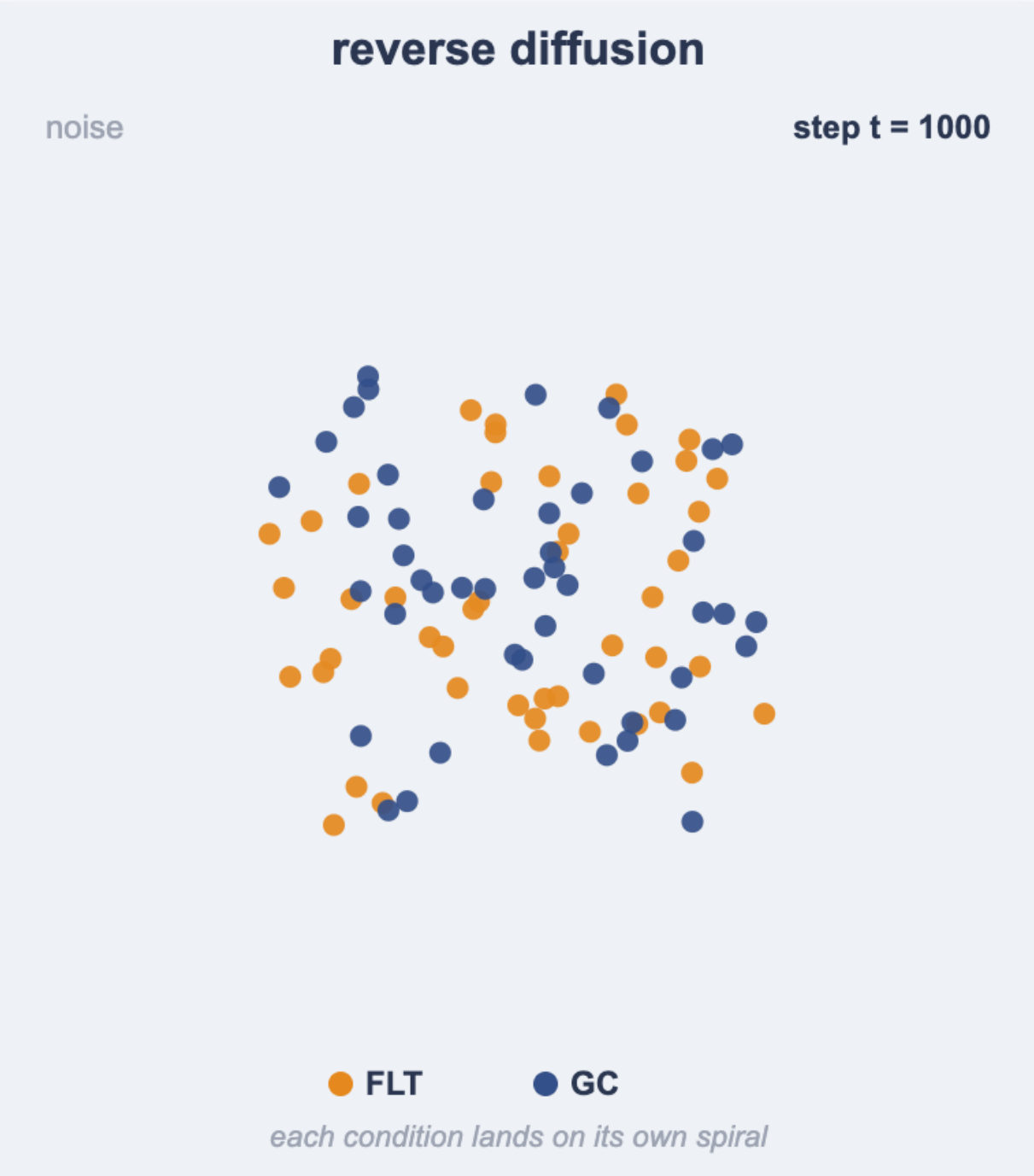
Diffusion — denoise from noise

Start from pure noise and denoise step by step onto the data manifold — conditioned on FLT or GC.

FAMILIAR · IMAGE DIFFUSION



OURS · GENE-EXPRESSION DIFFUSION

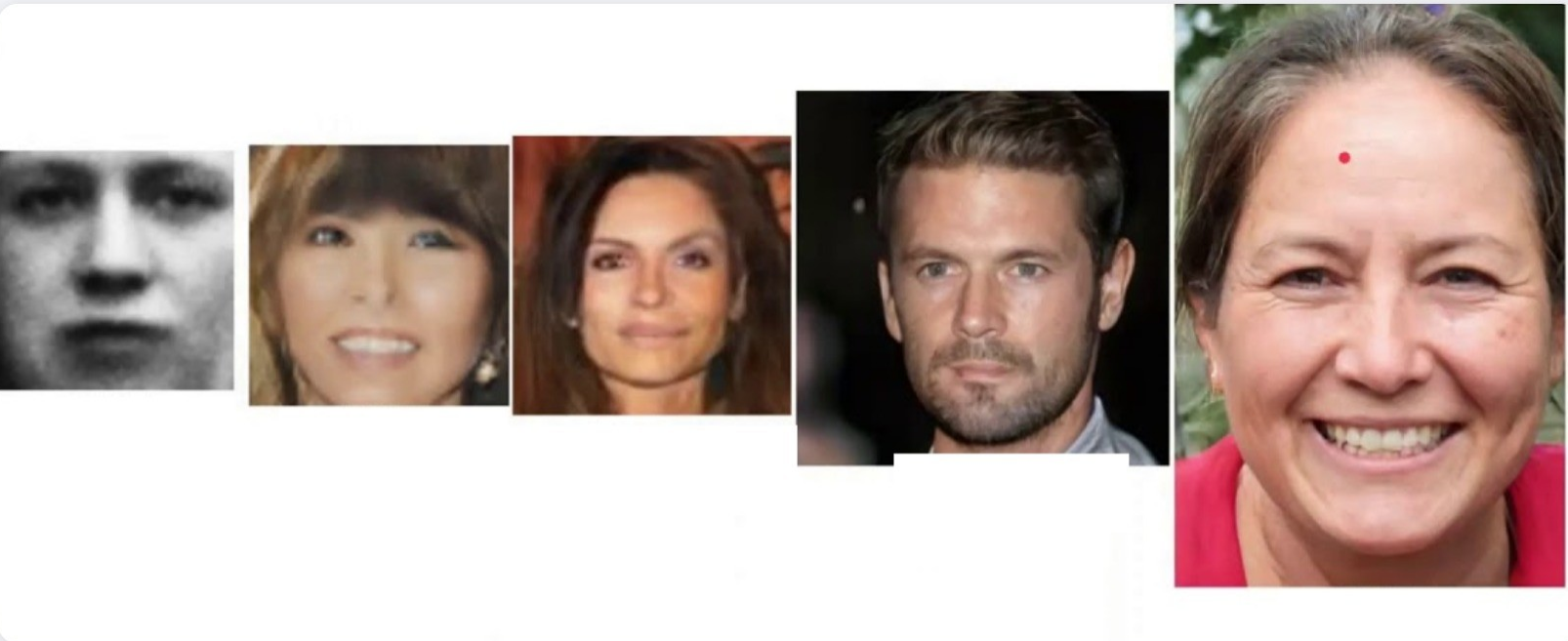


HOW THE GENERATORS WORK · 2 OF 3

Conditional WGAN-GP — generator vs critic

A generator and a Wasserstein critic compete until fake profiles are indistinguishable from real.

FAMILIAR · GANS FOR FACES



OURS · PROFILE GENERATOR



HOW THE GENERATORS WORK · 3 OF 3

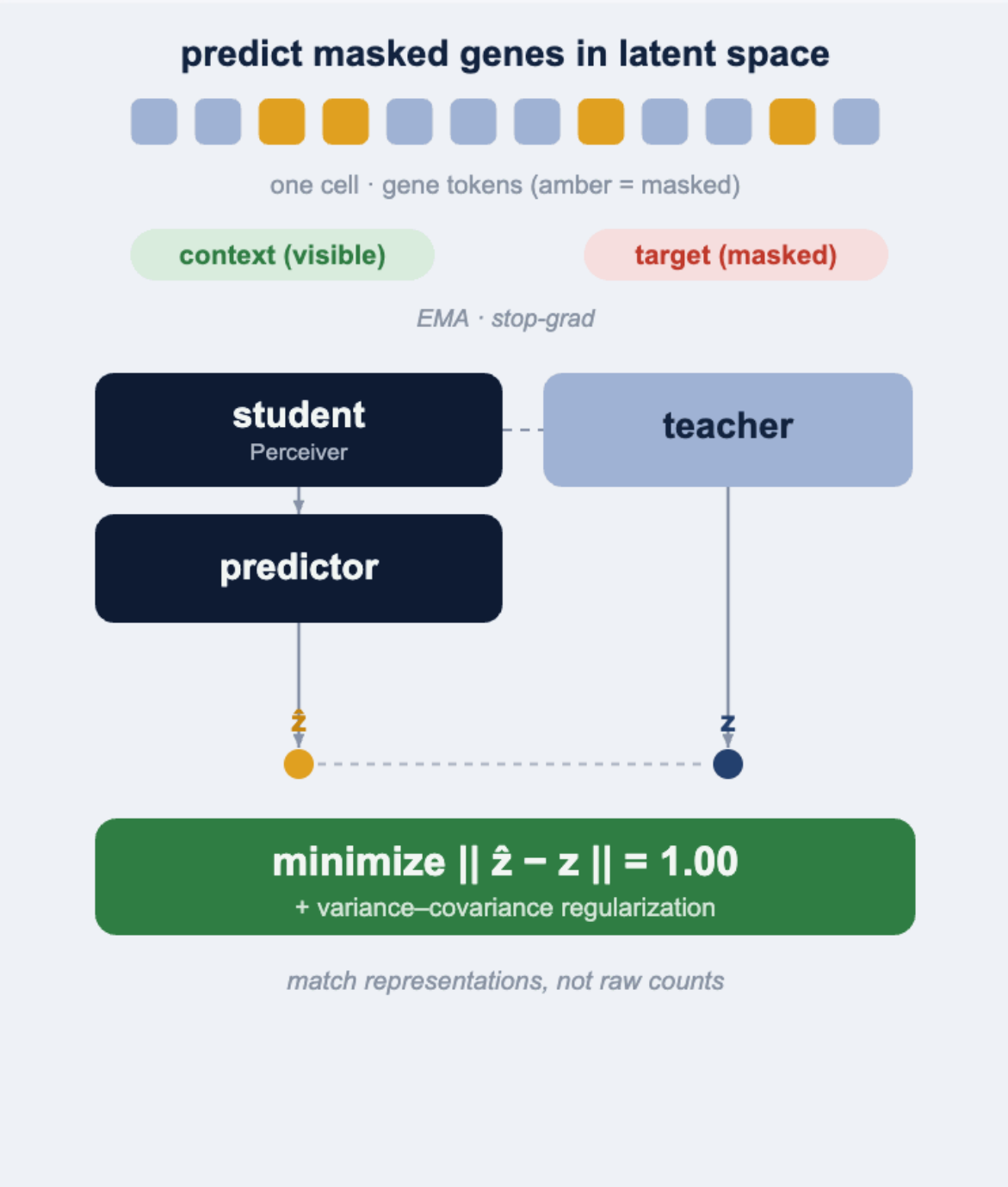
GeneJEPA — predict masked latents

Predict the latent representation of masked gene sets — a foundation model to build on.

FAMILIAR · I-JEPA FOR IMAGES



OURS · GENEJEPA

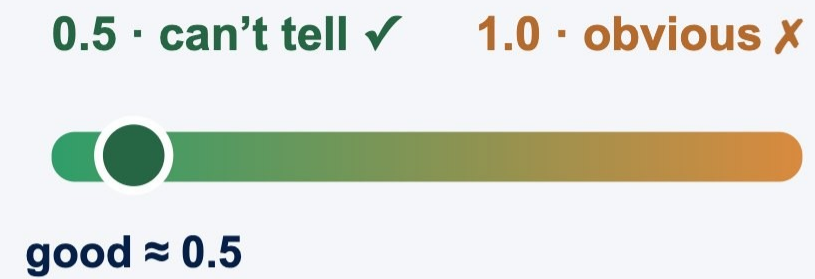


VALIDATION

Is the synthetic data any good?

Can a classifier tell them apart?

train a real-vs-synthetic classifier



Do DE results agree?

run DGEA on real and on synthetic



ACKNOWLEDGEMENTS

Thank you

MENTOR	James Casaletto
PROGRAM & DATA	NASA Space Life Sciences Training Program NASA Open Science Data Repository (OSDR) / GeneLab & contributing studies
OPEN-SOURCE METHODS	<u>GLARE</u> · <u>expiMap</u> · <u>MOBER</u>
AI ASSISTANCE	ChatGPT · Claude